

ISRAT JAHAN TULIN

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AI RESEARCHER

AI consultant and machine learning engineer with a strong background in machine learning, deep learning, computer vision, and scientific data analysis. Currently working at HZDR in the AI consultant team, focusing on self-supervised denoising of ROFEX sinograms acquired from a specialized HZDR imaging system. My work involves **CAREamics**, **Noise2Void**, **N2V2**, and **Structured N2V** for sinogram **denoising**, followed by tomographic reconstruction using denoised data. Over the past year, I have developed practical experience in reproducible AI workflows, including **Git**; **testing**; **MLOps** best practices such as **CI/CD**, **containers**, and **DVC** pipelines; **FAIR** data principles; secure data handling; **Unix/Linux shell scripting**; and environment management with **Conda**, **uv**, and **Pixi**. I also have extensive experience with **HPC** systems and scalable scientific computing and development environments such as **PyCharm** and **Visual Studio Code**. In addition, I contributed to open-source scientific software by identifying an early-stopping issue in **CAREamics/N2V** workflows and proposing a solution. My broader expertise includes **biomedical image analysis**; deep learning for **medical applications** such as pancreas tumor **classification**, **image segmentation**, **object detection**, **prediction**, and **feature extraction**; **sensor** and **time-series data analysis**; **multimodal data integration**; **data preprocessing**; and **PCA** and **ETL pipelines** using **Python**, **TensorFlow**, **Keras**, **Scikit-learn**, **Pandas**, **NumPy**, **SciPy**, **Matplotlib**, **TensorBoard**, and **PyTorch**. I am also active in **teaching** and scientific communication, having delivered machine learning and deep learning lectures at HIDA, the HPC Gateway Fall School, and the ErUM Data Hub and presented at events including Brains on Silicon and Building Bridges. I enjoy working in interdisciplinary teams and translating complex scientific problems into robust, reproducible, and practical AI solutions.

PROFESSIONAL EXPERIENCE

Helmholtz Zentrum Dresden Rossendorf (HZDR)

April 2025 - Present

Wissenschaftliche Mitarbeiterin (Machine Learning Engineer)

- Machine learning research for scientific imaging, including ROFEX sinogram denoising with CAREamics.
- AI consulting, teaching, and training for interdisciplinary research teams.
- Proposal writing, including an accepted **Helmholtz AI** funding application.
- Upcoming anomaly detection project for scientific datasets with ROTOP Pharmaka GmbH.
- Scientific writing as a [paper](#) co-author, including a **HAICON 2026** accepted [poster](#) abstract.
- Conference speaker/presenter for the ongoing tomography denoising project.

Otto-von-Guericke-University Magdeburg (OVGU)

Feb 2025 - March 2025t

Wissenschaftliche Mitarbeiterin (Machine Learning Engineer)

- Contributed to an AI-based interactive spine application using patient sensor data

Technische Universität Dresden (CeTI)

April 2021 - Jan 2025

Wissenschaftliche Hilfskraft mit abgeschlossener wissenschaftlicher Hochschulbildung

- Developed robotics simulations in Unreal Engine and Unity, leveraging C#, C++, VR/AR tools, Leap Motion, motion capture systems, and haptic devices for hand motion replication and collision-based feedback.
- Retrieved sensor data from motion capture (Qualisys, QTM Manager) and B&K sensors, measured latency to enhance precision, foster human-machine integration, and advance the Internet of Skills.
- Upgraded the [FingerTac](#) wearable device from single to three-finger, integrating it into Unreal Engine in collaboration with [Dr. Thomas Hulin](#) from DLR, to improve augmented haptic feedback.
- Collaborated with cross-functional teams in an agile environment to integrate quality benchmarks, optimize data pipelines, deploy models, and refine project goals for seamless system integration.

Technische Universität Dresden (ABSOLUT project)

January 2020 - June 2020

Wissenschaftliche Hilfskraft mit abgeschlossener wissenschaftlicher Hochschulbildung

- Researched ML-based object detection for safer autonomous transport systems.
- Developed stochastic occupancy grids and RANSAC-based detection for improved vehicle safety.
- Cleaned, transformed, and integrated stereo image data for accurate model training.
- Focused on data preprocessing, data transformation, and model validation for accurate object detection.

BUFT, SMUCT, Gono University

January 2014 - September 2019

Lecturer (Bangladesh)

- Delivered lectures on C, Python, operating systems, databases, and machine learning.
- Supervised student thesis projects, providing mentorship and guidance.
- Promoted interdisciplinary collaboration through cross-departmental projects and participated in professional conferences to stay updated on cutting-edge research and trends in machine learning.
- Published in peer-reviewed journals: [Artikel 1](#), [Artikel 2](#), [Artikel 3](#)

SOFT SKILLS

- Strong problem-solving, communication, and collaboration skills.
- Quality-focused and goal-oriented, with a proactive, agile mindset toward learning, adaptability, and continuous improvement.
- Organized and structured working style, with the ability to work independently and in teams.
- Proficient in English (C1) and German (B2.1).

EDUCATION

Master of Science in Computational Modeling and Simulation (<i>Technische Universität Dresden</i>)	2019-2024
Focus (Visual Computing): Machine Learning, Computer Vision, Data Science, High-Performance Computing	
Master of Science in Computer Science and Engineering (<i>Jahangirnagar University, Bangladesh</i>)	2012-2013
Bachelor of Science with Honors in Computer Science and Engineering (<i>Jahangirnagar University, Bangladesh</i>)	2009-2012

PROJECTS

Self-supervised Denoising Of Raw Tomography Detector Data For Improved Image Reconstruction	2025/26
- Researching self-supervised deep learning for ROFEX sinogram and raw tomography detector data denoising. - Evaluating denoising performance using PSNR, SSIM, CTF, MTF, and reconstruction-quality analysis. - Applying test cases and validation workflows for ongoing tomography denoising experiments. - Using Git, GitLab, and CI pipelines to sync CollabText and HedgeDoc documentation with GitLab. - Co-authored an accepted Helmholtz AI/AIRE funding proposal and contributed to an accepted HAICON 2026 poster abstract. - Co-authoring a scientific paper on the ongoing tomography denoising work. - Supporting AI teaching and training through ERUM Hub, HPC Gateway Fall School, and HIDA. - Presented sinogram denoising research at Building Bridges and Brains on Silicon conferences. - Working extensively with HPC for deep learning experiments and scientific workflows. - Preparing to lead an upcoming anomaly detection project with ROTOP Pharmaka GmbH.	
AI-based interactive spine application using patient sensor data	2025
Supported an AI-based Spine Interactive app for spinal disorders, using smartphone accelerometer and gyroscope data for mobility, gait-parameter analysis, and cognitive test integration.	
Predict Baltic Sea Temperature and Salinity using two years of 6-hourly sampled data.	2024
- Enhance predictions using the Baltic Sea's real-life temporal and spatial data, leveraging time series data processing to surpass naive baselines (e.g., persistence, linear regression) with an ML model (LSTM + CNN). Technology Used: Python, PyCharm, scikit-learn, TensorFlow, CI/CD, DVC, Unit Testing, Container (Docker)	
Deep-learning-based quantification of pancreatic texture using IR-Spectroscopy data	2023/24
- Developed a deep learning system using IR spectroscopy, real-life data collected from 30 patients at a university hospital, to analyze pancreatic tissue textures and tumor grades, aiding surgeons in assessing pancreatic cancer and post-operative fistula risk, under Prof. Dr. Stefanie Speidel at NCT, Dresden. - Cleaned and integrated data from multiple sources to build robust classification datasets, using scratch-built and pre-trained ResNet18 models and MobileNetV1, achieving over 90% accuracy with ResNet18. - Enhanced performance by integrating multimodal data (IR spectroscopic images and tumor grades) and applying dimensionality reduction techniques such as PCA to address dataset limitations, quality issues, and class imbalance. - Applied findings from leading journals to develop methodologies, optimize algorithms for improved model performance, accuracy, and interpretability, and collaborated with oncologists on clinical applications. - Implemented MLOps best practices by using GitLab for version control and continuous integration, DVC for model versioning, CI/CD for automated workflows, and TensorBoard with PyTorch for drift analysis when introducing new datasets, ensuring seamless model management, reproducibility, and monitoring. Technology used: Python, NumPy, Pandas, Matplotlib, PyTorch, Captum, PCA, TensorFlow, Keras, PyCharm, ScikitLearn, CI/CD pipelines, and Git, applying both supervised and semi-supervised learning techniques.	
Understanding Deep-Learning requires rethinking generalization	2020/21
- Explored deep neural network architectures' superior generalization compared to traditional regularization techniques, conducting experiments with AlexNet, VGG19, and ResNet. - Achieved near-zero training errors with noisy data, with performance drops on shuffled training/testing data and random labels. - Demonstrated the impact of regularization (early stopping, batch normalization) and optimization techniques (quantization, pruning) on generalization, emphasizing the need for new strategies to manage generalization errors. Technology used: Python, TensorFlow, Keras, JupyterLab, ScikitLearn.	
Exploring Medical Data with AR and Spatially Tracked Tablet	2020
- Developed a Unity3D simulation in C# for HoloLens and Tablet interaction with 3D liver CT images. - Enabled volumetric data creation, spatial navigation, and gesture-based interaction for enhanced organ visualization. - Developed HoloLens-based gesture interaction with a liver model, enabling selection of specific CT regions for operation planning.	
Kidney Segmentation in Abdominal CT Images Using Fuzzy Logic and Region Growing	2017
- Developed a fuzzy logic-based filtering technique to enhance abdominal CT images and preserve edges. - Applied region-growing algorithm for kidney segmentation, refining with ROI identification, pixel filling, erosion, labeling, and dilation. - Achieved 73% accuracy and a PSNR of 26.09 dB, outperforming traditional methods in quality and speed.	
Object-Oriented Programming with Java	2022
- Gained object-oriented programming skills through the SCPROG course at TU Dresden, focusing on Java. - Completed projects on core Java OOP concepts, including inheritance, polymorphism, interfaces, and generics. - Collaborated with peers to enhance problem-solving and debugging skills. - Utilized GitLab for version control, improving project management and teamwork.	